MOBILE COMPUTING

ASSIGNMENT 2 – CO2

MOBILE TELECOMMUNICATION SYSTEMS AND MULTIPLE ACCESS TECHNIQUES

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Course: Mobile Computing

ASSIGNMENT OVERVIEW

This assignment studies mobile communication and mobile computing, along with the multiple-access methods SDMA, TDMA, FDMA, CDMA and OFDMA. It also examines practical deployment decisions, simulation-based evaluation and redesign of a system when its original approach becomes inefficient.

The report follows the sequence required by the assignment: predict, design, implement, measure, analyse, evaluate, diagnose, redesign and validate. Numerical values that are not supplied in the question are stated as modelling assumptions. Simulation results are not invented; places requiring actual execution are identified clearly.

Learning coverage

Area / Main topics covered

- Q1 — Main topics covered: Mobile communication versus mobile computing; mobility and connectivity classification
- Q2 — Main topics covered: FDMA and TDMA channel sharing, overhead and traffic behaviour
- Q3 — Main topics covered: DS-CDMA spreading, despreading, processing gain and near-far problem
- Q4 — Main topics covered: OFDMA subcarrier orthogonality, allocation, throughput and fairness
- Q5 — Main topics covered: SDMA antenna arrays, beamforming and angular separation
- Q6 — Main topics covered: Weighted comparison and selection of access techniques
- Q7 — Main topics covered: Diagnose a bottleneck, redesign, implement and validate

Important modelling note

The handout gives some system parameters but leaves several quantities to be selected by the student, such as guard overhead, traffic datasets, SIR threshold, minimum user rate, simulation length and scoring weights. The report uses illustrative assumptions to show a complete engineering method. These assumptions should be replaced or confirmed using the actual course requirements and cited technical references before final submission.

Q1 – MOBILE COMMUNICATION OR MOBILE COMPUTING?

1.1 Introduction

Wireless communication means that information is transferred without a physical communication cable over at least part of the path. Mobile communication focuses on maintaining communication while a user or device changes location. Mobile computing is broader: it combines a portable computing device, wireless or intermittent connectivity, software and data processing so that users can perform computing tasks while moving.

Therefore, wireless and mobile are not identical. A fixed wireless router may use radio communication but remain installed at one location. A laptop moved between offices and connected through Ethernet is nomadic, even though its current connection is wired. Mobile

computing must also handle battery limits, changing networks, disconnections, storage and synchronisation.

1.2 Classification criteria

Criterion / Meaning / Reason for using it

- User mobility — Meaning: Whether the user/device moves during normal operation; Reason for using it: Separates fixed equipment from portable and mobile equipment
- Connection medium — Meaning: Wired, wireless or a mixture; Reason for using it: Identifies whether radio access is involved
- Disconnection tolerance — Meaning: Whether the application can continue while offline; Reason for using it: Important for intermittent coverage
- Location awareness — Meaning: Whether the application uses current location or context; Reason for using it: Useful for field-service and navigation applications
- Local processing and storage — Meaning: Whether work can be performed on the device; Reason for using it: Distinguishes a computing application from a simple communication terminal

1.3 Classification of the five cases

Case / Classification / Main problem area / Reason

- Rugged tablet fetching work orders over cellular data — Classification: Mobile; Main problem area: Mobile computing and mobile communication; Reason: The staff member moves, uses cellular connectivity and runs a work-order application.
- Laptop connected through Ethernet at different offices — Classification: Nomadic; Main problem area: Mainly mobile computing; Reason: The device changes location between sessions, but uses a wired connection at each office.
- Fixed wireless router at remote substation — Classification: Fixed-wireless; Main problem area: Mainly mobile communication / wireless access; Reason: The radio link is wireless, but the router and endpoint are fixed.
- Push-to-talk handset — Classification: Mobile; Main problem area: Mainly mobile communication; Reason: It provides voice communication while the user moves.
- Meter-reading tablet working offline and synchronising later — Classification: Mobile; Main problem area: Mobile computing with intermittent communication; Reason: The app stores readings locally and synchronises when a connection returns.

1.4 Additional ambiguous case

Example: a delivery vehicle contains a permanently installed computer connected to a cellular modem. The vehicle moves, but the computer is fixed inside the vehicle. It can be classified as mobile at the network level because its radio connection changes cells, while the terminal is fixed relative to the vehicle. The application is a mobile-computing system if it processes routes, stores delivery records and synchronises data.

1.5 Store-and-forward simulation design

The handout asks for link availability of 90%, 60% and 30%, comparing immediate transmission with a store-and-forward queue. For a reproducible experiment, assume 1,000 meter updates, one transmission attempt per update, and independent link availability for each attempt. An update is delayed if the first attempt fails but later succeeds; it is lost only if the simulation ends before successful delivery.

Link availability / Expected immediate successes / Expected immediate failures

- 90% — Expected immediate successes: 900; Expected immediate failures: 100
- 60% — Expected immediate successes: 600; Expected immediate failures: 400
- 30% — Expected immediate successes: 300; Expected immediate failures: 700

For the store-and-forward method, updates are placed in a local queue when the link is unavailable. The queue retries when connectivity returns. If the experiment includes enough retry opportunities and the queue does not overflow, eventual delivery can approach 100%. The exact delayed count and mean delay depend on the chosen retry schedule and simulation duration, so they must be measured rather than assumed.

1.6 Example simulation pseudocode

```
SET random_seed
FOR availability IN [0.90, 0.60, 0.30]:
    immediate_delivered = 0
    queued_delivered = 0
    delayed = 0
    lost = 0
    FOR each update in 1..1000:
        IF random() < availability:
            immediate_delivered += 1
            queued_delivered += 1
        ELSE:
            lost += 1
            add update to local_queue
    RETRY queued updates during later time slots
    count successful retries and delivery delays
    REPORT delivered, delayed, lost, mean_delay
```

1.7 Analysis and conclusion

The store-and-forward approach is more suitable for offline meter readings because it separates data collection from immediate network availability. It requires application-level changes such as local storage, timestamps, retry logic, duplicate prevention, queue limits and conflict handling. A wireless link alone cannot provide these functions. The rugged tablet and offline meter-reading application need strong mobile-computing support, while the push-to-talk handset mainly depends on reliable mobile communication.

Q2 – SHARING THE CHANNEL: FDMA VS TDMA UNDER DIFFERENT TRAFFIC

2.1 Given system

The assignment specifies a 2 MHz band per direction. FDMA uses 25 kHz channels. TDMA uses 200 kHz carriers, each containing eight time slots. Both are described as offering 80 nominal channels.

Nominal capacity is not necessarily usable capacity. Guard bands, guard times, synchronisation, control signalling and other overhead reduce the resources available for user traffic.

2.2 Nominal capacity calculation

For FDMA, 2,000 kHz / 25 kHz = 80 nominal channels. For TDMA, 2,000 kHz / 200 kHz = 10 carriers; 10 carriers × 8 time slots = 80 nominal time-slot channels.

2.3 Illustrative overhead assumptions

The handout asks the student to state and justify overhead values. For an example comparison, assume FDMA loses 5% of the band to guard and control requirements. Assume TDMA loses 1 of every 8 slots on each carrier to control and synchronisation overhead. These are modelling assumptions for comparison, not universal standards values.

Scheme / Nominal resources / Illustrative overhead / Estimated usable resources

- FDMA — Nominal resources: 80 channels; Illustrative overhead: 5%; Estimated usable resources: 80 × 0.95 = 76 channel-equivalents
- TDMA — Nominal resources: 80 time slots; Illustrative overhead: 1 slot in 8 per carrier; Estimated usable resources: 10 carriers × 7 traffic slots = 70 traffic slots

2.4 Traffic conditions

Traffic dataset / Traffic pattern / Expected FDMA behaviour / Expected TDMA behaviour

- A – Continuous voice — Traffic pattern: Long, steady calls; Expected FDMA behaviour: Dedicated channels are suitable; predictable allocation; Expected TDMA behaviour: Slots are regularly used; timing and synchronisation are required
- B – Bursty data — Traffic pattern: Short reports; about 10% duty cycle; Expected FDMA behaviour: A dedicated channel remains reserved while idle; Expected TDMA behaviour: Time slots can be shared or scheduled, reducing idle resource time
- C – Mixed traffic — Traffic pattern: 70% voice and 30% data; Expected FDMA behaviour: Voice channels are predictable, but data bursts may waste reserved capacity; Expected TDMA behaviour: Scheduling can allocate slots according to demand, but delay and control overhead must be considered
- D – Edge case — Traffic pattern: Many short calls arrive at nearly the same time; Expected FDMA behaviour: Call setup and channel allocation may create blocking peaks; Expected TDMA behaviour: Slot scheduling may queue bursts and create short-term congestion

2.5 Utilisation and blocking measures

Utilisation can be measured as the fraction of available traffic resources carrying useful payload. Blocking is the fraction of call or transmission requests that cannot obtain a resource immediately. Wasted capacity is reserved resource time that carries no useful payload.

For continuous voice, a dedicated FDMA channel can be efficient because the call uses the channel for much of its duration. For low-duty-cycle telemetry, dedicated allocation is usually inefficient because a device may reserve a channel while sending only a short report. A shared scheduled resource is generally more appropriate.

2.6 Campus telemetry design decision

At the telemetry site, 90% of devices send a 200-byte report per minute. If each device receives a dedicated FDMA or TDMA channel, most of the reserved resource may remain idle between reports. A shared access and scheduled transmission design is preferable, provided that collision handling, delay limits and device synchronisation are properly designed.

A final numerical decision requires the number of devices, packet overhead, available transmission interval and acceptable latency. These values are not supplied in the handout and should be stated as assumptions in the implementation.

Q3 – CDMA: ORTHOGONALITY, SPREADING AND THE NEAR-FAR PROBLEM

3.1 CDMA principle

Code Division Multiple Access allows multiple users to transmit in the same frequency band at the same time. Each user is assigned a spreading code. The transmitter multiplies each data bit by the code chips, increasing the signal bandwidth. The receiver uses the matching code to despread the desired user's signal.

3.2 Spreading and despreading

For a simple bipolar representation, data bit b is +1 or −1 and a spreading code contains chips c_1, c_2, …, c_L, each +1 or −1. The transmitted chip sequence is $x_i = b \times c_i$. At the receiver, the received chips are multiplied by the desired code and summed. A positive sum indicates bit +1; a negative sum indicates bit −1.

The processing gain is approximately Gp = L, where L is the spreading-code length. In decibels, $Gp(dB) = 10\log_{10}(L)$.

Code length L / Linear processing gain / Processing gain (dB)

- 8 — Linear processing gain: 9.03 dB
- 16 — Linear processing gain: 12.04 dB
- 64 — Linear processing gain: 18.06 dB

3.3 Orthogonality and correlation

Two codes are orthogonal when their inner product over the aligned code interval is zero. Orthogonal Walsh codes can separate synchronous users under ideal timing and channel conditions. Random non-orthogonal codes generally have non-zero cross-correlation, so one user's signal can interfere with another.

3.4 Predicted behaviour

Condition / Prediction as user count increases / Reason

- Synchronous orthogonal codes with equal power — Prediction as user count increases: Low multi-user interference in the ideal model until code resources or noise limits are reached; Reason: Aligned orthogonal codes have zero cross-correlation
- Random non-orthogonal codes — Prediction as user count increases: Bit errors generally increase with the number of users; Reason: Non-zero cross-correlation creates multiple-access interference
- Orthogonal codes with one user at 10× power — Prediction as user count increases: Weak users may experience errors even if codes are nominally orthogonal; Reason: Power imbalance and imperfect channel conditions create a near-far problem

3.5 Near-far problem and power control

The near-far problem occurs when a nearby handset reaches the base station with much greater power than a distant handset. The strong signal can dominate the receiver and make the weaker user's data difficult to detect. Power control adjusts handset transmit power so that received powers are more balanced.

Orthogonality can also be lost when users are not synchronised, when multipath propagation introduces delayed copies, or when channel conditions distort the codes. Therefore, ideal code orthogonality alone does not guarantee error-free reception.

3.6 Simulation plan

1. Generate random data bits for several simultaneous users.
2. Create Walsh orthogonal codes and random non-orthogonal codes at lengths 8, 16 and 64.
3. Spread each user's data and combine the transmitted signals at the receiver.
4. Apply the selected power levels and optional timing offsets.
5. Despread each user with its assigned code and make a bit decision.
6. Count bit errors and calculate BER = number of bit errors / total transmitted bits.
7. Repeat for increasing user counts and compare the three required conditions.

The supported number of simultaneous users should be reported as the largest tested user count that meets a chosen BER target. The BER target must be stated by the student before running the experiment. The assignment does not prescribe a particular BER threshold.

Q4 – OFDMA: GIVING EACH USER THE SUBCARRIERS THEY CAN USE BEST

4.1 OFDMA principle

Orthogonal Frequency Division Multiple Access divides a wide channel into many narrow, mutually orthogonal subcarriers. Different groups of subcarriers can be assigned to different users. Since each user's channel quality varies with frequency, the scheduler can allocate subcarriers according to user demand and channel gain.

4.2 Subcarrier spacing and symbol duration

For a basic OFDM system, the useful symbol duration T is the reciprocal of subcarrier spacing Δf: $T = 1/\Delta f$. Orthogonality is maintained when the subcarrier spacing and symbol timing are selected consistently.

A cyclic prefix copies the end portion of an OFDM symbol and places it at the beginning. It helps absorb multipath delay spread and reduces inter-symbol interference when the channel delay spread is within the cyclic-prefix duration. The cyclic prefix adds overhead and reduces the fraction of time carrying new data.

4.3 Numerical orthogonality check

For subcarriers with integer frequency indices k and m over a useful symbol interval T, the normalised inner product is proportional to the integral of $\exp(j2\pi(k-m)t/T)$ over the interval. It equals 1 when $k = m$ and 0 when $k \neq m$. A numerical implementation can sample the waveforms and calculate the inner product for at least three subcarriers.

4.4 Allocation approaches

Approach / Method / Expected advantage / Possible limitation

- Fixed contiguous blocks — Method: Each user receives a neighbouring block of subcarriers; Expected advantage: Simple scheduling and predictable allocation; Possible limitation: A user may receive weak subcarriers even when better ones exist elsewhere
- Interleaved round-robin — Method: Subcarriers are distributed in a repeating pattern among users; Expected advantage: Spreads frequency diversity across users; Possible limitation: Does not necessarily match each user's best channel conditions
- Channel-aware best-subcarrier — Method: Assign each subcarrier to the user with the best channel gain or utility; Expected advantage: Can increase total throughput; Possible limitation: Weak users may receive few resources and fairness can decline

4.5 Throughput and fairness

A simplified subcarrier throughput model is $R_{u,k} = B_k \log_2(1 + SINR_{u,k})$, where B_k is the subcarrier bandwidth and SINR is the signal-to-interference-plus-noise ratio for user u on subcarrier k. Total throughput is the sum of the allocated user rates.

Jain's fairness index is $J = (\Sigma R_u)^2 / [U \times \Sigma(R_u^2)]$, where U is the number of users. The index approaches 1 when users receive similar rates and becomes smaller when the rates are highly unequal.

4.6 Lecture-hall decision

For a lecture hall, a pure maximum-throughput scheduler may repeatedly allocate the best subcarriers to users with favourable channel conditions. This can starve users with weaker channels. A proportional-fair or minimum-rate-constrained scheduler is more suitable when every student must receive a minimum rate.

The minimum rate must be chosen from the application requirement. The handout does not specify it. A defensible implementation should define a minimum rate, verify whether total available capacity can meet it, and then maximise remaining throughput without violating the minimum.

Q5 – SDMA: REUSING THE SAME CHANNEL IN DIFFERENT DIRECTIONS

5.1 SDMA principle

Space Division Multiple Access separates users by their spatial direction. A base station with directional antennas or an antenna array can form a beam toward one user while reducing radiation toward other directions. Users at sufficiently different angles may reuse the same time and frequency resource.

5.2 Uniform linear array model

For an N-element uniform linear array with element spacing d, wavelength λ and steering angle θ_0, a normalised array factor can be written as $AF(\theta) = (1/N) \times \Sigma$ from n=0 to N−1 of $\exp[j\, n\, (2\pi d/\lambda)(\sin\theta - \sin\theta_0)]$. The magnitude of this expression describes the relative beam pattern.

Increasing N generally produces a narrower main lobe and improves angular selectivity, assuming suitable element spacing and calibration. Side-lobe behaviour depends on array weighting; a uniform amplitude array has a characteristic side-lobe pattern.

5.3 Required pattern evaluation

Array elements / Expected main-lobe behaviour / Engineering interpretation

- N = 2 — Expected main-lobe behaviour: Broad beam; Engineering interpretation: Limited ability to separate nearby users
- N = 4 — Expected main-lobe behaviour: Narrower than N = 2; Engineering interpretation: Improved directional separation
- N = 8 — Expected main-lobe behaviour: Further narrower beam; Engineering interpretation: Better spatial selectivity, with more RF and processing complexity
- N = 16 — Expected main-lobe behaviour: Narrowest among listed arrays; Engineering interpretation: Higher potential separation but more calibration and hardware complexity

The assignment asks for beam patterns at user separations from 5° to 60°. Exact beamwidth, side-lobe level and SIR values depend on the element spacing, wavelength, steering convention and array weights. A common illustrative choice is $d = \lambda/2$ with uniform element weights; the student should state this before computing the patterns.

5.4 SIR and angular separation

When the beam is steered toward user A, the desired gain is evaluated at θA and the interfering gain at θB. A simplified SIR at user B or at the receiver can be formed from the desired and interfering received powers, including path loss and beam gains. The minimum angular separation is the smallest separation for which the chosen SIR threshold is met.

The handout does not prescribe the SIR threshold. An illustrative threshold such as 10 dB may be selected for the experiment, but it must be justified and used consistently.

5.5 Practical limitations and toll-plaza decision

SDMA becomes difficult when users have similar arrival angles, when angular spread is large, or when multipath causes signals to arrive from several directions. Two users arriving from almost the same direction may produce highly overlapping beams and cannot be reliably separated by direction alone.

At a toll plaza, most vehicles arrive from the same direction. SDMA alone is therefore not a dependable separation method. It may still help distinguish lanes or users with different angular positions, but time, frequency or code scheduling should provide the main resource separation.

Q6 – CHOOSING MULTIPLE-ACCESS TECHNIQUES FOR REAL DEPLOYMENTS

6.1 Evaluation criteria and weights

A weighted decision matrix makes the selection transparent. The following illustrative weights add to 100%. A score of 1 means poor suitability and 5 means very good suitability. The weights and scores are design judgements and should be reviewed against the evidence from Q2–Q5.

Criterion / Weight / Reason

- Capacity / spectral efficiency — Weight: 25%; Reason: Important when many users share limited spectrum
- Robustness to interference — Weight: 20%; Reason: Maintains service in crowded radio environments
- Delay and responsiveness — Weight: 15%; Reason: Important for voice and interactive applications
- Fairness / access predictability — Weight: 15%; Reason: Prevents users or devices from being starved
- Implementation complexity — Weight: 10%; Reason: Affects cost and operational difficulty
- Scalability for many devices — Weight: 15%; Reason: Important as the network grows

6.2 Deployment recommendations

Deployment / Primary technique / Possible hybrid / Reason

• Wide-area cellular voice — Primary technique: TDMA / FDMA combination; Possible hybrid: FDMA carriers with TDMA slots; Reason: Provides structured channel and time-slot allocation for voice traffic.

• Jam-resistant tactical link — Primary technique: CDMA; Possible hybrid: Frequency hopping or directional antenna support where permitted; Reason: Spreading can provide processing gain and interference resilience; performance depends on synchronisation and power control.

• High-rate stadium downlink — Primary technique: OFDMA; Possible hybrid: OFDMA with SDMA beamforming; Reason: Subcarrier scheduling supports high data rates and beamforming can reuse resources spatially.

• 5,000 low-rate sensors — Primary technique: Scheduled TDMA or shared access; Possible hybrid: Low-duty-cycle scheduling with aggregation; Reason: Avoids reserving a permanently dedicated channel for every low-rate sensor.

6.3 Why systems combine access techniques

Real systems combine methods because each technique addresses a different dimension. FDMA separates frequency bands, TDMA separates transmission times, CDMA separates users by spreading codes, OFDMA allocates orthogonal subcarriers and SDMA separates users spatially. A hybrid system can use more than one dimension to improve capacity and manage interference.

6.4 Sensitivity analysis

To perform the required ±20% sensitivity check, multiply each criterion weight by 0.8 and 1.2 in separate trials, then renormalise all weights to 100%. Recalculate each technique's weighted score. If the preferred method changes, the choice is sensitive to the weighting. If it remains the same, the decision is more robust.

The final score table should be completed using the student's own Q2–Q5 experimental evidence. The handout explicitly requires ratings to be traceable to that evidence; therefore, this report does not present invented simulation scores.

Q7 – DAY-4 DESIGN CHALLENGE: DIAGNOSE, REDESIGN AND VALIDATE

7.1 Selected limitation

Selected case: OFDMA channel-aware best-subcarrier allocation may maximise total throughput but can starve users with weak channel conditions. This is one of the limitations listed in the assignment. It is suitable for redesign because the original channel gains can be reused to design a fairer allocation without replacing the entire radio technology.

A. Diagnose the bottleneck

In best-subcarrier allocation, the scheduler repeatedly assigns a subcarrier to the user with the highest channel gain. Users with consistently weak channel gains may receive very few subcarriers. The counted quantity to monitor is each user's allocated subcarrier count and achieved throughput, along with the total sum throughput and Jain's fairness index.

The limitation is caused by maximising instantaneous channel quality without a minimum service guarantee. The bottleneck is not necessarily a lack of total spectrum; it can be an allocation policy that repeatedly favours users with strong channels.

B. Define the design objective

The redesign objective is to prevent starvation while retaining as much total throughput as possible. A measurable objective is: every active user must receive at least one scheduling opportunity within a defined scheduling window, and the scheduler should maximise throughput subject to a selected minimum-rate constraint.

C. Design the revised strategy

Use a fairness-aware scheduler. First, check which users have not met their minimum service requirement. Give those users priority for suitable available subcarriers. Then allocate remaining subcarriers using a proportional-fair score that balances channel quality against the user's recent average throughput.

This method reuses channel-gain measurements and recent throughput information. It adds a small amount of state for each user, such as average throughput and minimum-rate status. The trade-off is that the total throughput may be slightly lower than a pure best-channel allocation.

Revised scheduling pseudocode

```
INPUT: channel_gain[user][subcarrier], average_rate[user]
INPUT: minimum_rate, available_subcarriers

FOR each scheduling interval:
  update each user's achieved_rate
  urgent_users = users below minimum service target

  FOR each subcarrier:
    IF urgent_users is not empty:
      choose eligible user with greatest service deficit
    ELSE:
      choose user maximising
        channel_rate(user, subcarrier) /
        max(average_rate[user], small_value)

    assign subcarrier to selected user
    update predicted and achieved user rates

  update average_rate for all users
  record sum throughput and Jain fairness
```

D. Predict before implementing

Prediction: the fairness-aware scheduler should increase the minimum user throughput and Jain's fairness index compared with pure best-subcarrier allocation. Its total throughput may decrease slightly because some subcarriers will be assigned to users whose channels are not the strongest.

E. Implement and validate

Use the same generated channel gains and random seed for both schedulers. Run at least 1,000 channel realisations as required in Q4. Record sum throughput, per-user throughput, minimum user rate, number of subcarriers per user and Jain's fairness index. The comparison must use identical channel conditions.

Measure / Original best-subcarrier / Redesigned fairness-aware

- Sum throughput — Original best-subcarrier: To be measured
- Minimum user throughput — Original best-subcarrier: To be measured
- Average user throughput — Original best-subcarrier: To be measured
- Jain's fairness index — Original best-subcarrier: To be measured
- Subcarrier allocation per user — Original best-subcarrier: To be measured

These are result fields, not measured values. They must be populated from the executed simulation. A valid conclusion should explain both improvements and trade-offs.

F. Defend the final design

If the redesigned scheduler improves minimum rate and fairness while retaining acceptable sum throughput, it is a better choice for a lecture hall where every student must receive service. Pure best-subcarrier allocation may still be preferable for short-term peak throughput when fairness is not a requirement or when all users already meet their minimum service targets.

7.2 Overall engineering conclusion

The redesign is successful only if the measured evidence supports the prediction. If throughput falls too much, the minimum-rate target or scheduling weights should be adjusted and tested again. The final decision must report the magnitude of change and identify any new computational or signalling cost.

CONCLUSION

This assignment compared mobile communication and mobile computing and examined five multiple-access techniques. FDMA and TDMA divide resources by frequency and time, CDMA separates users by spreading codes, OFDMA allocates orthogonal subcarriers and SDMA separates users by spatial direction. Each technique has strengths and limitations that depend on traffic type, interference, mobility, fairness and implementation complexity.

For practical deployments, a single access technique is not always sufficient. Hybrid methods can combine the benefits of frequency, time, code, subcarrier and spatial separation. The Day-4 redesign demonstrates that a system should be improved by first identifying its actual bottleneck and then validating a measurable change under equivalent test conditions.

REFERENCES

8. ITA03-MP, Mobile Computing, Assignment 2 – CO2, supplied course handout (10 pages).

9. Rappaport, T. S., Wireless Communications: Principles and Practice, 2nd Edition, Prentice Hall, 2002.
10. Goldsmith, A., Wireless Communications, Cambridge University Press, 2005.
11. Proakis, J. G. and Salehi, M., Digital Communications, 5th Edition, McGraw-Hill, 2008.
12. Stallings, W., Wireless Communications and Networks, 2nd Edition, Pearson, 2005.

APPENDIX A – FORMULAS AND PERFORMANCE METRICS

Concept / Formula / definition / Use

• FDMA nominal channels — Formula / definition: Total bandwidth / channel bandwidth; Use: Estimate frequency-channel count

• TDMA nominal channels — Formula / definition: Number of carriers × time slots per carrier; Use: Estimate time-slot resources

• Processing gain — Formula / definition: $Gp = L$; $Gp(dB) = 10\log_{10}(L)$; Use: Measure spreading benefit

• Bit error rate — Formula / definition: BER = bit errors / total transmitted bits; Use: Evaluate CDMA reliability

• OFDM useful symbol duration — Formula / definition: $T = 1/\Delta f$; Use: Relate subcarrier spacing and symbol duration

• OFDMA user rate — Formula / definition: $R_{u,k} = B_k \log_2(1 + SINR_{u,k})$; Use: Estimate per-subcarrier throughput

• Jain fairness index — Formula / definition: $J = (\Sigma R_u)^2 / [U\Sigma R_u^2]$; Use: Measure throughput fairness

• Array factor — Formula / definition: $AF(\theta) = (1/N)\Sigma\exp[jn(2\pi d/\lambda)(\sin\theta - \sin\theta_0)]$; Use: Evaluate beam directionality

APPENDIX B – EXPERIMENTAL DATA CHECKLIST

- Record the random seed, number of users, number of bits or channel realisations and all selected assumptions.
- Keep the test conditions identical when comparing original and redesigned methods.
- Save raw data in CSV format and retain the source code used to generate it.
- Report counted quantities such as bits, errors, slots, blocked requests, subcarriers and throughput.
- Include labelled tables and graphs; state units on every axis.
- Do not describe predicted values as measured results.
- Explain unexpected results and identify whether they arise from the model, assumptions or implementation.

APPENDIX C – GLOSSARY

Term / Meaning

• Mobile communication — Meaning: Communication maintained with a user or device that may change location.

• Mobile computing — Meaning: Computing using portable devices, applications and data while moving or intermittently connected.

• FDMA — Meaning: Frequency Division Multiple Access; users receive separate frequency channels.

• TDMA — Meaning: Time Division Multiple Access; users share a carrier in different time slots.

• CDMA — Meaning: Code Division Multiple Access; users are separated using spreading codes.

• OFDMA — Meaning: Orthogonal Frequency Division Multiple Access; users receive different orthogonal subcarriers.

• SDMA — Meaning: Space Division Multiple Access; users are separated by spatial direction or beams.

• Spreading code — Meaning: A chip sequence used to spread a user's data signal.

• Near-far problem — Meaning: A strong nearby signal interferes with reception of a weaker distant user's signal.

• Cyclic prefix — Meaning: A copied portion of an OFDM symbol added to reduce multipath-induced inter-symbol interference.

• Beamforming — Meaning: Controlling antenna phases and amplitudes to direct radio energy.

• Jain's index — Meaning: A numerical measure of fairness among users' achieved rates.

APPENDIX D – SUBMISSION REQUIREMENTS

The course handout requires a digital assignment report in PDF format containing evidence for Q1–Q7, source code for all implementations and simulations, a GitHub repository link with a README and fixed random seeds, raw experimental data in CSV format, and the tables and graphs used in the analysis.

The handout also states that counted quantities are primary experimental evidence. Execution time may be included as supporting evidence but must not be the only basis for conclusions. Standards parameters such as frame durations, rates and header sizes must be obtained from cited references, and assumed values must be clearly identified.

For assessment verification or viva, the student should be able to explain the design, code, test-data design, predictions, experimental results and conclusions. Any AI or reference tools used should be disclosed as required by the course handout.